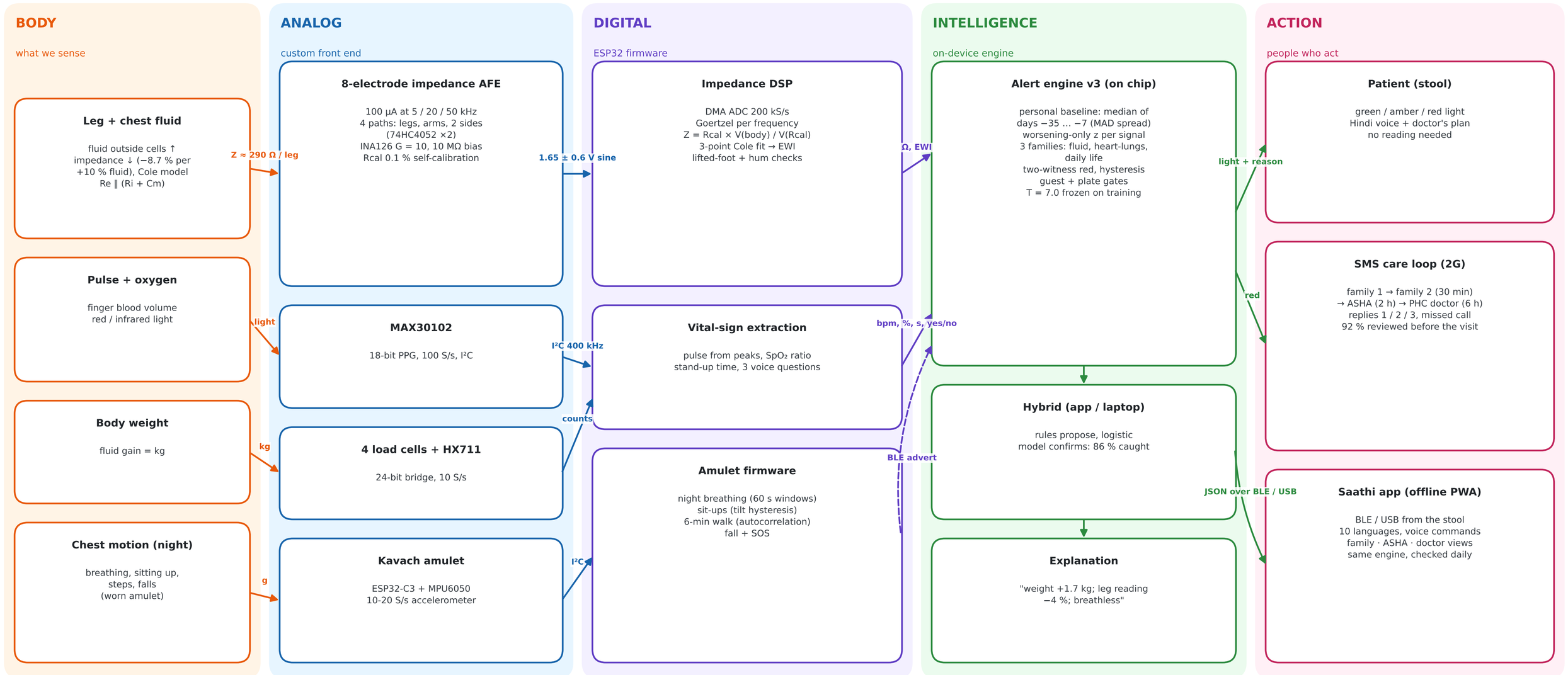
HridayMitra: system architecture, from the body to the doctor

Each arrow is labelled with what really flows through it (values from LTspice, the firmware emulator and the synthetic cohorts). Home unit parts ₹3,917-4,820. Monitoring, not diagnosis.



SAFETY, POWER AND FAILURES (designed in, each one tested)

Patient safety

IEC 60601-1 type BF: every single fault ≥ 12.8× under limit; C3 + C4 block DC

CHARGE / USE switch

cuts both charger wires: leakage 0.19 µA (339 µA with a cheap charger)

Power cut / no network

18650 battery; SMS queue retries every 10 min; engine state in flash

Bad contact / guest

lifted foot named + re-measured; toe ring > 25 % = no reading; guest not saved

Part failure

Rcal ratio checked at each check-in: outside 2.0-2.4 → no numbers + service SMS

Pacemaker / ICD

never on the plates: weight-only mode, no magnets in the amulet